Embedding Strategy

KJV 1611 / KJV 1769 Semantic Drift Study

Purpose

This document formalizes the embedding strategy for the study. It defines the embedding model, platform, named vector architecture, point identity rules, vector normalization assumptions, distance metric, and provenance requirements that make vector comparisons interpretable.

Embedding Model Specification

All vectors used in the study are generated with mxbai-embed-large through the Ollama platform. The same model and platform configuration are used for KJV 1611, KJV 1769, R0, and R1. Model mixing across canons, representations, or runs is not permitted.

Embedding Consistency Rules

The study requires model uniformity, inference consistency, and text-to-vector mapping integrity. Each R0 vector must correspond to text_raw, and each R1 vector must correspond to text_norm. Cross-mapping between representations is prohibited.

Named Vector Architecture

The Qdrant architecture uses a single collection with named vectors. Each point contains the same payload and deterministic point identity while storing separate representation-specific vector fields.

Representation	Vector Name	Source Field
R0	v_text_raw	text_raw
R1	v_text_norm	text_norm
R2	v_text_norm_experiment	Reserved future experimental normalization

Point Identity Contract

Each Qdrant point is uniquely associated with canon_id, work_code, division_level_1, and division_level_2. This identity contract ensures that R0 and R1 vectors correspond to the same text unit and that cross canon pairing is deterministic.

Vector Normalization and Distance Metric

Qdrant normalizes vectors under cosine configuration. The study therefore treats vectors as unit vectors in embedding space. Cosine similarity is equivalent to the vector dot product, and cosine distance is defined as one minus cosine similarity.

- $\text{drift}_i = 1 - \text{cosine\_similarity}(v_{1611}, v_{1769})$

- drift_i = 1 - ($v_{\_1611}$ dot $v_{\_1769}$) when vectors are unit normalized

Embedding Provenance Requirements

Each vector is traceable to its embed_model, model_id, canon_id, representation type, stable unit identifier, and unit checksum. These fields support reproducibility, auditability, and future reruns of the same study design.

Study Contract

Component	Study Decision
Embedding model	mxbai-embed-large
Platform	Ollama
Vector structure	Named vectors in one Qdrant collection
R0 vector	v_text_raw
R1 vector	v_text_norm
Distance metric	Cosine distance
Vector normalization	Unit-normalized vectors under cosine configuration

Expected Output Artifacts

- manifest_run.json
- vector_index_state records
- drift_pairs.csv